INCH-POUND
MIL-I-60292A
30 March 1990
SUPERSEDING
MIL-I-60292(MU)
7 July 1966 and
Amendment 1,
2 November 1966
and Notice 1, 1986

MILITARY SPECIFICATION

INITIATOR, CARTRIDGE ACTUATED, M99 ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the M99 Cartridge Actuated Initiator Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-2550	Ammunition, General Specification For
MIL-I-60525	Initiator, Cartridge Actuated, M99 Metal Parts Assembly
MIL-I-82767	Initiator, Cartridge Actuated, M99 Remanufactured Metal Parts Assembly

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-453	Inspection, Radiographic
MIL-STD-1168	Ammunition Lot Numbering
MS28741	Hose Assembly, Detachable End Fitting, Medium Pressure

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1377

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those cited in the solicitation (see 6.2).

DRAWINGS

NAVAL ORDNANCE STATION, INDIAN HEAD, MD (CAGE CODE 19200 and 19203)

8597230	Cartridge, Impulse, M91 Assembly
DL10520405	Initiator, Cartridge Actuated, M99, Assembly
8830984	Box, Fiberboard (Inner) Packing, Ammunition for Initiator, Cartridge Actuated M27, M86, M87, M98, M99, or XM107
8830985	Box, Packing, Ammunition for Initiator, Cartridge Actuated M27, M86, M87, M98, M99, or XM107
9241155	Box, Fiberboard (Outer) Alternative, Packing Ammunition for Initiator, Cartridge Actuated, M27, M86, M87, M98, M99, or XM107

U.S. AIR FORCE

TPO 00-845-1059	Initiator, Cartridge Actuated, M99

(Application for copies should be addressed to: Commanding Officer, Naval Ordnance Station, Data Control Branch (Code 3720), Indian head, MD 20640-5000.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. All materials used in the manufacture of the initiators shall conform strictly to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.2 Primary components. For the purpose of this specification, the M91 delay cartridge (Drawing 8597230) is defined as a primary component. Only primary components from a single lot shall be used in a production lot of initiators. One primary component production lot may be used in more than one initiator production lot.

3.3 Breakaway torque. The chamber and cap of the assembled initiator shall withstand a breakaway torque application of 400 inch pounds without loosening or deformation (see 4.2.3.1).

3.4 Residual magnetism. The initiator shall not deflect the indicator of the compass more than five degrees in either direction using procedure 4.2.3.2.

3.5 Firing pin release. The firing pin of the assembled initiator, less safety pin, cartridge and chamber shall release from the cocked position when the application of an axial load of 18 to 25 pounds has withdrawn the initiator pin 0.370 inch, minimum, and 0.425 inch, maximum (see 4.2.3.3).

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3.6 Initiator pin security. Application of an axial load of 200 to 220 pounds to the initiator pin of the assembled initiator shall not withdraw the initiator pin past the safety pin. Load application shall cause no permanent deformation of the safety pin (see 4.2.3.4).

3.7 Functional (see 4.2.3.5).

3.7.1 Ballistics. Initiators loaded with cartridges from one accepted lot and temperature conditioned shall perform within the pressure values and ignition delay time (see 6.6.2) specified in table I. Pressure values shall be recorded at the end of 15 feet of aircraft hose, assembly number MS28741-4, and a pressure chamber with a volume of 1 cubic inch.

TABLE I. *Pressure values.*

Temperature (°F)	Pressure (psi)		*Ignition delay time, max (ms)* (see 6.6.2)
	Minimum	Maximum	
−60 to −70	500	1425	65
65 to 75	600	1550	65
195 to 205	700	1700	65

3.7.2 Misfires. There shall be no misfires.

3.7.3 Mechanical failures. The initiator shall function without mechanical failure (see 6.6.1).

3.8 Workmanship. Initiators shall be free of cracks, splits, incomplete or damaged threads on inlet and outlet ends, or other defects which might prevent proper mating with connecting parts or proper functioning of the device.

3.8.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

3.8.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.8.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

3.9 Radiographic examination. Each initiator assembly shall be radiographically examined for defects as specified in 4.4.

3.10 Component parts. All component parts shall meet the requirements of MIL-I-60525 or MIL-I-82767 prior to assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements (see 6.2).

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Quality assurance terms and definitions. Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.3 Inspection. Inspection shall be in accordance with MIL-A-2550.

4.2 Inspection provisions.

4.2.1 Lot.

4.2.1.1 Submission of product. Unless otherwise specified by the contracting officer, formation of lots and presentation of lots shall be in accordance with the "Submission of Product" provisions of MIL-STD-105.

4.2.1.2 Size of lot. A lot shall consist of an identifiable group of assembled initiators containing metal parts assemblies produced by a homogeneous process and loaded with one lot of accepted cartridges.

4.2.2 Examination. One-hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed in accordance with the classification of defects. Four power (4X) magnification may be used for inspection of such defects as burr and foreign matter. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic shall be measured or gaged to determine acceptability. All nonconforming material shall be rejected.

4.2.2.1 Cartridge, Impulse Delay, M91, Drawing 8597230, covering a detail of Drawing 10520405.

Characteristic	Defect	Method of Inspection
Critical		
1	Mixed cartridges	Visual
2	Metal defective	Visual
3	Marking missing, incorrect, or illegible	Visual
Major		
101	Foreign matter present	Visual
Minor		
201	Case scratched, wrinkled, bulged, or dented	Visual
202	Varnish missing or inadequate	Visual

4.2.2.2 Assembly, Drawing 10520405, less all drawings except 10520410, 10520411, and MS 19060-3.

Critical		
1	Protrusion of firing pin	Gage
2	Steel ball missing	Gage
3	Striker hole in firing pin housing obstructed	Visual

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Characteristic	Defect	Method of Inspection
Major		
101	Burr	Visual
102	Foreign matter present	Visual
Minor		
None defined		

4.2.2.3 Assembly, Drawing 10520405, less Drawings 8597230, 10520407, 10520496, 10520414, and MIL-C-5501/11.

Critical		
1	Firing pin not cocked	Visual/Manual
2	Any parts missing	Visual
3	Improper assembly	Visual
Major		
101	Safety pin loose or deformed	Visual
102	Burr	Visual
103	Foreign matter present	Visual
Minor		
None defined		

4.2.2.4 Assembly, Drawing 10520405, less all Drawings, except 10520407, 10520414, and 8597230.

Critical		
1	Cartridges not properly seated	Visual
2	Filter missing or insecure	Visual
3	Nylon patch missing or improperly applied	Visual
Major		
101	Burr	Visual
102	Foreign matter present	Visual
Minor		
None defined		

4.2.2.5 Assembly, Drawing 10520405.

Critical		
1	Cartridge face not seated against base of firing pin housing	X-ray
2	Any part missing except as otherwise classified	Visual/X-ray
3	Improper assembly	Visual/X-ray
4	Marking missing, incomplete, or illegible	Visual
Major		
101	Set screw missing or insecure	Visual
102	Burr	Visual
103	Foreign matter present	Visual

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Characteristic	Defect	Method of Inspection
Minor		
201	Thread sealing compound missing from set screws	Visual
202	Protective cap missing	Visual
203	Protective finish inadequate or defective	Visual

4.2.2.6 Box, Fiberboard (Inner) Ammunition Packing Drawing 8830984.

Critical

None defined

Major

None defined

Minor		
201	Contents loose	Manual
202	Box damaged	Visual
203	Sealing strip missing, improperly applied	Visual
204	Marking missing, incorrect, or illegible	Visual
205	Cut, damaged, or improperly sealed bag	Visual

4.2.2.7 Box, Ammunition Packing, Drawing 8830985.

Critical

None defined

Major

None defined

Minor		
201	Contents loose	Manual
202	Strapping missing, loose, broken, or improperly assembled	Manual/Visual
203	Contents of box exposed	Visual
204	Box damaged	Visual
205	Marking missing, incorrect, or illegible	Visual
206	Number of units, incorrect	Visual

4.2.2.8 Box, Fiberboard (Outer) Alternative Ammunition Packing, Drawing 9241155.

Critical

None defined

Major

None defined

Minor		
201	Contents loose	Manual
202	Box broken or damaged	Visual
203	Number of units contained incorrect	Visual
204	Marking missing, incorrect, or illegible	Visual
205	Tape incomplete	Visual

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4.2.3 Inspection testing. The following tests shall be performed in accordance with the provisions of 4.3.

4.2.3.1 Breakaway torque. Each assembly shall be subjected to the breakaway torque test of the threaded joint between the cap and the chamber. Failure of any assembly to comply with the requirements of 3.3 shall be cause for rejection of the initiator.

4.2.3.2 Residual magnetism. Each initiator shall be subjected to the test for excessive residual magnetism as specified in 4.3.2. Failure of any initiator to comply with the requirement of 3.4 shall be cause for rejection of the initiator.

4.2.3.3 Firing pin release. The initiator pin of each subassembly, consisting of all parts, except the safety pin, cartridge, and chamber, shall be subjected to the firing-pin release test. Failure of any subassembly to comply with the requirements of 3.5 shall be cause for rejection of the subassembly. To achieve the pull force requirement of 18-25 pounds specified in 3.5 selective reassembly may be necessary (i.e., changing initiator pins or springs) for those assemblies which are out of range.

4.2.3.4 Initiator pin security. The initiator pin of each subassembly consisting of all parts, except the chamber and cartridge, shall be subjected to the initiator-pin security test. Failure of any subassembly to comply with the requirement of 3.6 shall be cause for rejection of the subassembly.

4.2.3.5 Functional tests. The following tests shall be performed by the Government.

4.2.3.5.1 Ballistics. Twenty-one units shall be tested to determine compliance with the requirements of 3.7. The units shall be temperature conditioned as provided in table II and 4.3.5. Failure of any unit to comply with the requirements of 3.7 shall be cause for rejection of the lot.

4.2.3.5.2 Retest. There shall be no retests.

4.2.3.5.3 Test failure. If a test failure is attributable to an assignable cause, excluding the test initiator, the original test results shall be discarded and that test reconducted.

TABLE II. *Conditioning temperature.*

Temperature (°F)	Sample size
−60 to −70	7
65 to 75	7
195 to 205	7

4.3 Test methods and procedures.

4.3.1 Breakaway torque. A torque wrench set at the torque requirement specified in 3.3 shall be used in conjunction with a fixture to determine the adequacy of the joints. The chamber shall be secured in an air vise and the torque shall be applied to the cap.

4.3.2 Residual magnetism. The assembly shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects by placing the assembly 5 inches from the end at the same height as the compass, in a north-south horizontal position (cap end at south) with the compass set in an east-west heading. The assembly shall be moved at a speed no greater than 10 feet per minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis, and moved slowly back past the compass. This procedure shall be repeated for each 90 degrees rotation of the assembly for 4 full passes.

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4.3.3 Firing pin release. The subassembly shall be secured in the fixture. A pull shall be applied to the initiator pin in the longitudinal direction, in conjunction with the fixture which registers the amount of pull.

4.3.4 Initiator pin security. The subassembly shall be secured in the fixture with the safety pin in place. A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with a fixture which registers the amount of pull. The test shall be performed with the safety pin in both positions.

4.3.5 Functional.

4.3.5.1 Ballistics. The test sample shall be rigidly supported in the fixture and fired. The sample shall be tested within 5 minutes after removal from the temperature conditioning chamber. When this time limit is exceeded, the sample units shall be reconditioned in accordance with 4.3.5.2. Required pressure-time readings shall be taken as specified in 3.7.1.

4.3.5.2 Temperature conditioning. Each assembled initiator shall be conditioned for a period of time not less than 4 hours nor greater than 24 hours. When reconditioning initiators at −60 °F, all condensation shall be removed from the interior surfaces before returning the initiator to the conditioning chamber.

4.4 Radiographic examination. All assembled initiators shall be radiographically examined in accordance with MIL-STD-453, with exception that an initiator assembly will be used for the penetrameter. Any observable imperfections in components or assembly shall be cause for rejection. The initiators shall be positioned for the most revealing exposure, with the long axis of each assembly perpendicular to the rays of the X-ray machine. All initators shall be identified with serial numbers prior to X-ray examination. The serial numbers shall be in consecutive numerical order beginning with the number 001 in each production lot. Any missing serial numbers are to be identified on the X-rays. The initiators shall be arranged on boards or trays in consecutive order and each radiograph shall carry a permanent identification of the units displayed thereon. The radiographic plate identification shall include item nomenclature, the complete lot number, the span of serial numbers displayed, and the contract number. The radiographs of the entire production lot shall accompany the ballistic sample to the activity conducting the production lot acceptance test. Defective initiators found during X-ray review are to be marked on the X-ray prior to shipment and those defective items shall be removed from the production lot. Film interpretation of the radiographs shall verify the presence and correct positioning of the following parts:

a. O-ring present	P/N 10520496
b. Steel balls present (3 ea)	P/N MS19060-3
c. Firing pin	P/N 10520411
d. Spring, initiator	P/N 10520412
e. Seal	P/N 10520415
f. Firing pin housing	P/N 10520410
g. Filter	P/N 10520414
h. Cartridge, Impulse M91	P/N 8597230
i. Initiator pin	P/N 10520409

4.5 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of section 5.

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5. PACKAGING

5.1 Packing and marking. Packing and marking shall be Levels A or B as specified by the contracting activity:

Level A - In accordance with Drawings 8830984 and 8830985
Level B - In accordance with Drawings 8830984 and 9241155

5.1.1 Lot numbers shall be in accordance with MIL-STD-1168.

5.2 Air Force packaging. Air Force packaging shall be in accordance with TPO 00-845-1059.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The initiator assembly is used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

6.2 *Acquisition requirements.* *Acquisition documents must specify the following:*

a. Title, number, and date of this specification
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).
c. Production test activity and lot size (see 4.1 and 4.2.1).
d. Direction for shipping radiographs of entire production lot along with ballistic sample to the activity conducting production lot acceptance tests (see 4.4).
e. Level of marking and packing (see 5.1)
f. Safety precautions (see 6.4).

6.3 Explosive hazard classification information.

Hazard Classification/Division: 1.4
Storage Compatibility Group: S
DOT Class: C
DOT Marking — Explosive Power Device — Class C — Handle Carefully — Keep Fire Away

6.4 Safety precautions. The safety precaution requirements of "Contractors' Safety Manual for Ammunition, Explosives, and Related Dangerous Material" (DOD 4145.26M) are applicable and should be specified in the contract as required by the Federal Acquisition Regulation (FAR) 23.3. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP 5) should be made applicable.

6.5 Hazard notice. The initiator described herein and certain of its components are flammable, explosive, or both. Consequently, they present hazards in manufacture, handling, storage, and shipment. The contractor should recognize these hazards and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling, and electrically-induced incidents.

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6.6 Definitions.

6.6.1 Mechanical failure. Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

6.6.2 Ignition delay time. Ignition delay time is defined as the time from primer impact to the start of pressure rise.

6.7 Subject term (keyword) listing.

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue because of the extensiveness of the changes.

Custodian:
Navy-OS
Air Force-99

Preparing activity:
Navy-OS
(Project 1377-946)

Review activity
Air Force-70

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